



TECHNICAL NOTE

VARIABILITY OF FEMORAL MUSCLE ATTACHMENTS

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Abstract—Analytical and experimental models of the musculoskeletal system often assume single values rather than ranges for anatomical input parameters. The hypothesis of the present study was that anatomical variability significantly influences the results of biomechanical analyses, specifically regarding the moment arms of the various thigh muscles.

Insertions and origins of muscles crossing or attaching to the femur were digitized in six specimens. Muscle volumes were measured; muscle attachment area and centroid location were computed. To demonstrate the influence of inter-individual anatomic variability on a mechanical modeling parameter, the corresponding range of muscle moment arms were calculated.

Standard deviations, as a percentage of the mean, were about 70% for attachment area and 80% for muscle volume and attachment centroid location. The resulting moment arms of the m. gluteus maximus and m. rectus femoris were especially sensitive to anatomical variations (SD 65%).

The results indicate that sensitivity to anatomical variations should be analyzed in any investigation simulating musculoskeletal interactions. To avoid misinterpretations, investigators should consider using several anatomical configurations rather than relying on a mean data set. Copyright © 1996 Elsevier Science Ltd.

Keywords: Anatomy; Femur; Muscular attachments; Inter-individual variability; Musculoskeletal modeling.

INTRODUCTION

The location and size of muscle attachments are important in several aspects for a detailed analysis of the mechanical behavior of long bones. On a large scale, muscles fulfill the functions of support, movement and performance in the skeleton. On a smaller scale, muscle forces provide appropriate stimulus for remodeling of the bony tissue. Because tendon insertions sometimes extend over large areas, mechanical analyses of muscle–bone interactions are sensitive to attachment geometry and location. Therefore, it is important to know the inter-individual anatomic variation of these attachments and to estimate the significance of this variability.

In many studies dealing with a wide range of biomechanical topics, muscle attachment locations have been quantified in lower limbs. Seireg and Arvikar (1973) estimated muscle attachment centroids from anatomical textbooks. Since anatomical views often are not to scale, are viewed from unknown directions and are two-dimensional, this method is of limited value. Crowninshield *et al.* (1978), Dostal and Andrews (1981), Pierrynowski and Morrison (1985) and White *et al.* (1989) located origin and insertion centroids of leg musculature on dry bone specimens. To the authors of this study it appeared difficult to exactly locate attachment sites and it was virtually impossible to describe attachment extents on dry bones without soft tissue remains. Brand *et al.* (1982) identified muscular attachment centroids in three pairs of intact lower limbs. Müller-Gerbl *et al.* (1990)

measured lower limb muscle coordinates on four specimens without reporting the measuring method. In a recent work Lengsfeld *et al.* (1994) used a mechanical device to digitize attachment locations on two specimens, but did not report any coordinate data or perform any comparisons.

In all studies mentioned muscle attachments were reduced to their area centroid. The difference between area centroid and muscle centroid, as described by An *et al.* (1981), is seldom discussed. Further, errors may have been introduced by the estimation of the centroid locations of larger attachment areas.

Anatomical variability has been described in only one study, using standard deviation as a measure of variation (Brand *et al.*, 1982). Osteometric scaling conceptually reduces standard deviations. However, White *et al.* (1989) reported large standard deviations for bony landmarks even after scaling. Therefore, it seems questionable, whether scaling techniques are always appropriate to reduce the differences in data sets to a mean anatomy.

In a different approach, imaging techniques were utilized to determine muscle attachments. As described by Chao *et al.* (1993, 1994), the three-dimensional geometry of muscles may be reconstructed from CT or MRI scans. Although the substantial amount of cross-sectional data allows adequate reconstruction of muscle attachments and centroid lines, the method is limited when the image plane is oriented parallel or at a small angle to the plane of muscle attachment (e.g. muscles in the hip region). The extensive resources required to analyze each subject has limited this technique thus far to only a few specimens and anatomical variations have not yet been reported.

The moment arms of muscles (Dostal *et al.*, 1986) are used in many aspects of musculo-skeletal modeling and have been intensively investigated (Inman, 1947; Olson *et al.*, 1972; Johnston *et al.*, 1979; Spoor *et al.*, 1990; Delp and Zajac, 1992). Because of their wide use in biomechanics, moment arms are considered

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good examples to illustrate the influence of inter-individual variability on musculo-skeletal analyses. The sensitivity of moment arms to other parameters has previously been discussed (straight vs centroid line modeling in An *et al.*, 1981, 1984; Jensen and Davy, 1975). A similar influence on muscle moment arms could be expected from inter-individual anatomical variability.

The purpose of this study was to determine the amount of variation in volume, attachment size and location of femoral muscles, and to investigate to sensitivity of moment arm calculations to this anatomical variation.

MATERIALS AND METHODS

Four male and two female specimens (lumbar spine L3 to L5 and lower left extremity), screened for adverse musculo-skeletal conditions, were evaluated (Table 1). Skin and fat were removed from the formalin-fixed specimens and markers were attached to 11 bony landmarks to determine the relative position of the bones (M1 to M11 in Fig. 1). To allow digitization by use of magnetic fields, the leg was fixed on a specially designed acrylic table (Fig. 2); the distance to any metal within the room exceeded 2 m. The table included several skeletal fixation devices to rigidly secure the leg in an orientation parallel to its long axis. The fixation devices were applied such that damage to muscles (e.g. vasti), muscle attachments (e.g. semitendinosus) and joint surfaces were avoided and the markers remained easily accessible. This fixation ensured that the pelvis, femur and lower leg were in the anatomical position assumed in a double leg stance. Specimens were protected against drying by wet towels wrapped around the leg.

Measurements were performed using the Polhemus 3Space Digitizing System (Polhemus, Colchester, Vermont) consisting of a magnetic field source, a digitizing pen and a control unit. The source was attached to the acrylic table close to the knee joint. Its orientation defined the global coordinate system (Fig. 1). By means of a foot switch, the three-dimensional position of each marker and the points of muscle attachment were measured in relation to the source position. According to the manufacturer, the static system accuracy is 0.8 mm RMS within a $477 \times 477 \times 265$ mm³ working volume. The specimens were

Table 1. Demographic data of specimens

Specimen	Age	Sex	Femur length (mm)	Tibia length (mm)
1	72	m	376	329
2	70	m	466	407
3	79	f	396	350
4	81	m	422	371
5	76	m	441	358
6	67	f	421	340

contained within this volume. Data collected from the 3Space was transmitted to a 386PC and stored in ASCII files for further calculations.

To check measurement repeatability, the 11 marker locations were digitized at the beginning and end of each measurement session and absolute differences were averaged. The anatomical structures were identified using anatomical textbooks (Clemente, 1987; Frick, 1987). The attachment areas of muscles crossing or inserting into the femur were digitized. To allow access to an attachment, the muscle-tendon complex was dissected from the bone. The number of points (minimum 20) and the digitization pattern were chosen to allow a computer-based reconstruction of the attachment boundary, the muscle centroid and the three-dimensional curvature of the area.

In some muscles, additional steps were necessary. Three different origin sites were identified for the mm. glutei. The m. gluteus maximus insertions into the iliotibial band and a proximal and distal aspect of the gluteal tuberosity were differentiated. The distal insertion led into the linea aspera via the lateral intermuscular septum. The iliotibial band was represented at its proximal end by the origin of the m. tensor fasciae latae and fibers of the m. gluteus maximus. Fibers from the m. gluteus medius inserting into the iliotibial band were not considered. At the distal end, the band was represented by insertions into the lateral patellar retinaculum and the lateral tibial condyle. Insertions of the iliotibial band via the lateral femur fasciae into the lateral intermuscular septum were not included. Mm. iliacus and psoas were handled as two independent muscles with identical

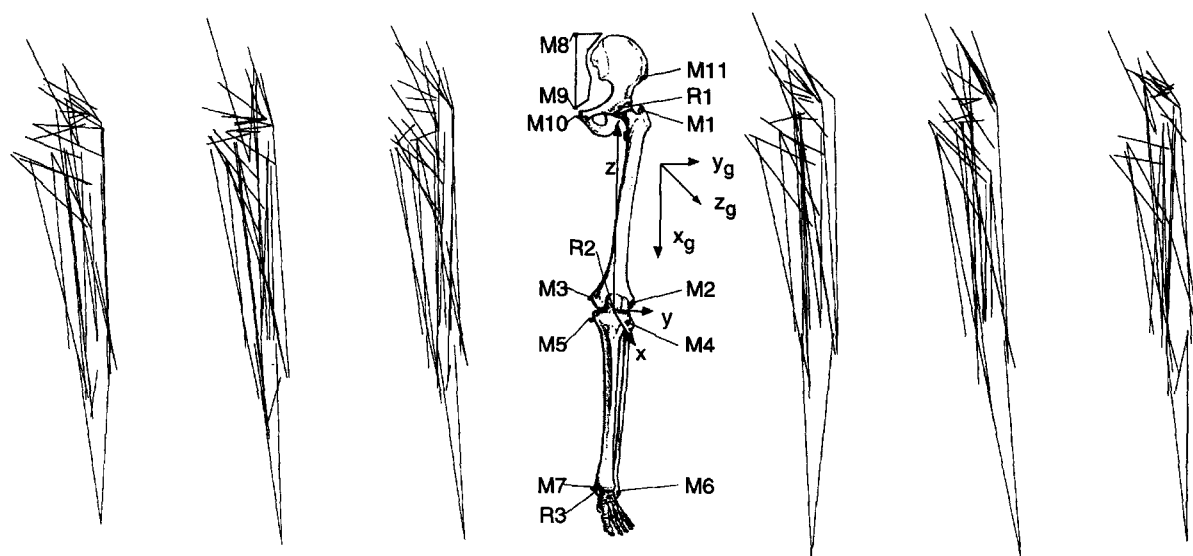


Fig. 1. Centroid-to-centroid straight line representation of muscles with femoral attachments in A-P view and lower extremity showing the location of marker and reference points. Femur-based (x, y, z) and global Cartesian (x_g, y_g, z_g) systems of coordinates.

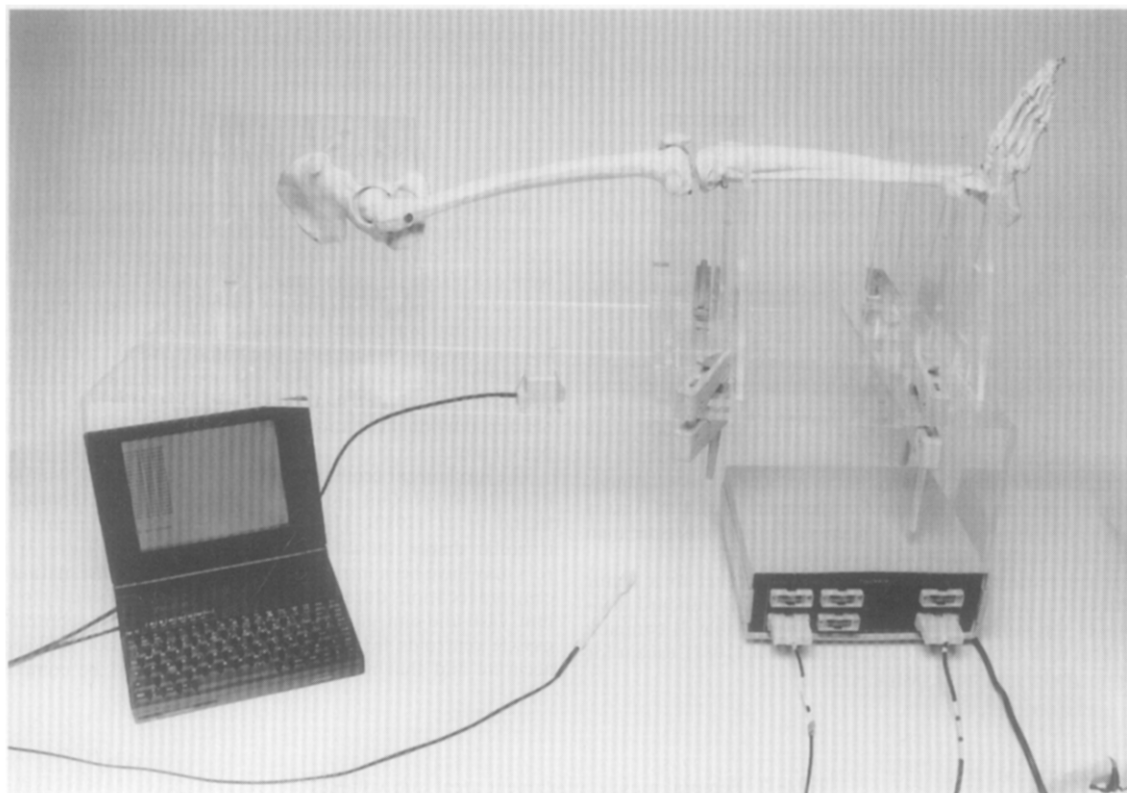


Fig. 2. Photograph of acrylic table, 3Space digitizer (source, pointer and control unit) and computer for data acquisition. Artificial bones are shown in place of specimens.

femoral insertions. For the m. adductor magnus, a cranial and a caudal insertion were digitized. If present, the m. adductor minimus was included. No other ligamentous structure, fascia or septum was included.

Femoral length was defined as the distance between the center of the femoral head (R1 in Fig. 1) and the centroid of the femoral insertion of the posterior cruciate ligament (R2). Tibia length was defined as the distance between R2 and the mid-point between lateral and medial malleoli (R3). The reference points (Fig. 1) were calculated based upon the following digitization protocol. To define the femoral head center, R1, the ventral part of the capsule was opened by a T-shaped cut so that parts of the femoral head surface were visible. The dorsal capsule of the hip joint was kept intact in order to secure the original position of the femoral head. Assuming the head to be a sphere, four non-planar points (three-dimensional vectors $\mathbf{p}_k$, $k = 1 \dots 4$) on the head surface were digitized three times and a mid-point vector ($\mathbf{m}$) and radius value (r) calculated.

$$|\mathbf{p}_k - \mathbf{m}| = r,$$

for vectors $k = 1 \dots 4$
and coordinates $j = 1 \dots 3$

$$\sum (p_k[j] - m[j])^2 = r^2.$$

The centroid of the femoral posterior cruciate ligament (PCL) insertion, R2, was calculated from 20 digitized points. Reference point R3 was calculated as the mid-point of lateral and medial malleolar markers (M6 and M7).

For specimens 3–6, the muscle volumes were determined using the volume displacement method described by Clauser *et al.* (1969).

The coordinate data of each specimen were transformed from the global into the femur-based system of coordinates (Fig. 1).

The origin of the femur-based system was reference point R2, the z-axis pointed toward reference point R1, and the x-axis was perpendicular to an axis through markers M2 and M3.

Using the surfacing tool of I-DEAS MasterSeries Version 1.3c (Structural Dynamics Research Corporation, Milford, Ohio) on a SGI Indy Workstation (Silicon Graphics Inc., Mountain View, California) the attachment areas could be reconstructed from the digitized points of the femoral muscle insertions. A free-face surface, e.g. a tensor product Bézier surface (Farin, 1990), was laid through all the points of an attachment site. The muscle attachment area size could then be calculated. Muscle attachments with the largest variance among the six specimens were used to quantify the influences of the computer-based reconstruction procedure and to check the repeatability of the method. These areas were digitized, reconstructed and calculated three times.

To better allow inter-specimen comparison of muscle attachment locations, osteometric scaling was performed based upon a least-squares solution (Sommer *et al.*, 1982) using a stretching, rotation and translation procedure. The reference points, the markers and the muscle centroid locations of the six specimens served as a database for the scaling procedure. All six specimens were scaled to a mean model (420 mm femur length). Then, the standard deviations were calculated within the group of six specimens for all scaled attachment centroids.

Moment arms were calculated for each individual muscle and for the joint spanned using the definition of Dostal *et al.* (1986). The center of the femoral head was assumed to be the hip center, R1, and the knee center was approximated by the centroid of the femoral PCL insertion, R2. To maintain the manageability of the data, only the moment arm magnitude is reported for each muscle. The moment arm components as well as the muscle attachment coordinates may be obtained from the authors.

RESULTS

The specimens did not show abnormalities from those described in standard anatomical textbooks (Clemente, 1987; Frick, 1987) except for the fact that a *m. adductor minimus* could only be located in one specimen (6). The *m. gemellus superior* in specimens 1 and 6, and the *m. gemellus inferior* in specimen 4 could not be identified. The *m. popliteus* could not be located in specimens 2 and 3. Problems in identifying the *mm. gemelli* or *m. popliteus* may have been due to their small size in the investigated specimens.

Repeatability of the marker position measurements was within a range of 2.5 ± 0.8 mm for all markers and specimens. Reproducibility of muscle attachment area calculation (Table 2) was checked in three trials for the muscles with the largest inter-individual anatomical variation: the *m. obturator internus* and the *m. vastus intermedius*. The inter-individual variation (SD 70%) was 10 times larger than the differences found in the reproducibility.

Total muscle volume was smallest in specimen 3, its *m. gluteus maximus* was largest. Except for specimen 5, the *m. vastus lateralis* was always larger than the *m. vastus medialis*. Specimen 4 had the greatest total muscle volume.

For comparison, the six data sets were scaled to a femur length of 420 mm (Fig. 1). The mean values and standard deviations of the muscle attachment locations (Table 3) compare favorable with those reported by Brand *et al.* (1982), except for the *mm. glutei* and *mm. vasti*.

The moment arms were calculated from the straight line muscle models displayed in Fig. 1 (Table 4). Large moment arm variations ($> 5\%$ femur length) were found for the *m. gluteus maximus* and *m. rectus femoris*.

DISCUSSION AND CONCLUSIONS

A digitizing method was developed which allows reproducible determination of muscular attachments as three-dimensional areas. Muscle volume, attachment area size and area centroid were calculated in six specimens. Previously unreported inter-individual muscle attachment variability far exceeded system accuracy and was found to be comparable to that for muscle volume and attachment size. The authors are not aware of any quantitative study on complete specimens with a larger sample size. Still, the reported anatomical variability gives only an idea of inter-individual variations generally possible within the whole population.

The mechanical centroid of muscle lines of action, computed from the attachment areas, were similar to those reported by Brand *et al.* (1982). However, comparing attachments with irregular shapes, the differences are pronounced (Table 3). Errors may have been introduced by Brand *et al.* by estimating the centroid locations during the masking process, especially for the *mm. vasti* and *mm. glutei*. It can be concluded, that determining muscle centroids from dry bone specimens is not sufficiently accurate and that attachment centroids should be computed

Table 2. Femoral muscle attachment areas and volumes

Muscles	Area (mm ²)						Volume (ml)			
	Specimen						Specimen			
	1	2	3	4	5	6	3	4	5	6
Gluteus maximus 1, 2	212	730	780	259	345	364	150	150	40	120
Gluteus maximus 3	1461	853	265	330	523	541	835	420	440	420
Gluteus medius 1, 2, 3	1246	960	1290	887	783	852	100	340	210	320
Gluteus minimus 1, 2, 3	685	487	530	519	305	503	60	50	80	30
Psoas major	143	616	413	448	137	130	20	70	110	50
Iliacus	143	616	413	448	137	130	50	120	90	70
Piriformis	95	393	86	183	254	80	15	20	20	20
Obturator internus	51, 50, 43	86	114	371, 366, 355	165	91	5	5	20	5
Obturator externus	101	146	145	166	157	77	30	30	10	40
Gemellus inferior	—	77	161	470	171	—	5	5	5	—
Gemellus superior	90	—	84	—	174	84	5	—	5	5
Quadratus femoris	669	733	732	450	209	123	40	30	40	20
Pectineus	53	112	429	197	145	230	40	50	20	40
Adductor breve	356	576	564	184	405	286	35	80	20	90
Adductor longus	324	492	736	449	326	276	40	90	60	90
Adductor minimus	—	—	—	—	—	101	—	—	—	40
Adductor magnus, pars cran.	819	1558	481	1006	374	440	165	380	220	240
Adductor magnus, pars cau.	61	155	167	218	189	241	55	190	80	60
Biceps femoris, caput breve	1093	880	821	859	733	887	40	100	50	60
Popliteus	26	—	—	159	166	202	—	5	5	5
Gastrocnemius lateralis	99	296	510	315	116	116	30	100	40	120
Gastrocnemius medialis	600	935	412	244	346	199	60	160	80	60
Vastus medialis	386	1136	522	192	221	198	120	260	300	300
Vastus intermedius	267	1214, 1150, 1203	532	149, 151, 167	316	748	105	140	200	180
Vastus lateralis	736	1393	1703	1054	866	970	230	380	250	460
Rectus femoris	—	—	—	—	—	—	80	140	100	90
Sartorius	—	—	—	—	—	—	45	110	50	110
Gracilis	—	—	—	—	—	—	40	130	50	70
Semitendinosus	—	—	—	—	—	—	40	160	90	130
Semimembranosus	—	—	—	—	—	—	75	180	150	190
Biceps femoris, caput longus	—	—	—	—	—	—	70	180	120	150
Tensor fasciae latae	—	—	—	—	—	—	50	80	40	80

Table 3. Scaled femoral muscle attachment centroids

Muscle name	Femoral muscle attachments (mm)											
	This study						Brand <i>et al.</i> (1982)					
	Mean			SD			Mean			SD		
	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
Gluteus maximus 1	10	46	380	16	13	41	-16	35	406	—	—	—
Gluteus maximus 2	0	38	322	13	13	27	-16	35	361	—	—	—
Gluteus maximus 3	5	44	375	7	15	43	-16	35	289	5	6	20
Gluteus medius 1, 2, 3	12	45	399	8	6	12	-20	60	390	6	13	37
Gluteus minimus 1, 2, 3	21	35	404	17	10	10	-7	57	381	7	5	45
Psoas major	5	9	357	7	7	11	-18	12	335	3	6	33
Iliacus	4	9	355	6	7	13	-18	12	335	3	6	34
Piriformis	1	31	417	11	9	10	-13	48	398	5	7	36
Obturator internus	-3	28	412	14	10	10	-11	45	395	12	7	32
Obturator externus	-2	12	396	16	22	13	-24	42	382	7	12	25
Gemellus inferior	-8	28	408	12	11	15	-11	45	395	12	7	32
Gemellus superior	-6	34	413	12	7	9	-11	45	395	11	7	32
Quadratus femoris	-10	28	381	13	12	21	-16	33	345	17	7	22
Pectineus	8	15	320	8	13	18	-11	25	315	13	3	32
Adductor breve	10	15	296	11	8	16	-8	22	283	11	2	58
Adductor longus	12	9	210	11	6	20	-3	13	192	3	4	20
Adductor minimus	11	21	298	—	—	—	-12	29	276	9	8	47
Adductor magnus, pars cran.	8	14	215	12	11	44	-4	16	174	1	4	64
Adductor magnus, pars cau.	-4	-33	25	9	11	14	-6	-30	17	6	3	4
Biceps femoris, caput breve	10	15	172	10	5	18	-1	14	178	5	7	0
Popliteus	-18	11	-14	10	12	7	—	—	—	—	—	—
Gastrocnemius lateralis	-16	25	19	9	10	15	-20	23	5	5	25	5
Gastrocnemius medialis	-15	-17	18	4	12	23	-20	-16	8	8	19	7
Vastus medialis	22	13	331	11	15	38	4	9	188	5	5	16
Vastus intermedius	28	26	321	14	13	50	23	18	207	3	11	23
Vastus lateralis	18	37	339	15	4	22	1	37	213	4	10	12

Table 4. Moment arms of femoral muscles to the hip and knee joint

		Moment arm magnitudes (mm)								
		Specimen						This study		Dostal <i>et al.</i>
	Joint	1	2	3	4	5	6	Mean	SD	(1986)
Gluteus maximus 1, 2	Hip	35	38	67	61	22	18	40	20	51
Gluteus maximus 3	Hip	41	39	100	86	37	22	54	31	—
Gluteus medius 1, 2, 3	Hip	33	41	47	38	51	46	43	6	71
Gluteus minimus 1, 2, 3	Hip	24	33	37	30	32	43	33	6	52
Psoas major/Iliacus	Hip	9	19	4	14	17	3	11	7	20
Piriformis	Hip	36	19	38	27	23	51	33	12	37
Obturator internus	Hip	30	26	48	27	16	35	30	11	33
Obturator externus	Hip	42	32	50	17	33	11	31	15	25
Gemellus inferior	Hip	—	26	54	36	15	—	33	17	34
Gemellus superior	Hip	39	—	39	—	25	37	34	6	31
Quadratus femoris	Hip	36	49	66	58	34	26	45	15	56
Pectineus	Hip	59	58	94	55	74	40	63	19	49
Adductor breve	Hip	87	85	59	84	85	73	79	11	79
Adductor longus	Hip	82	98	98	83	86	107	93	10	82
Adductor minimus	Hip	—	—	—	—	—	73	73	—	73
Adductor magnus, pars cran.	Hip	43	82	82	83	47	54	65	19	67
Adductor magnus, pars cau.	Hip	32	66	68	59	34	48	51	16	77
Biceps femoris, caput breve	Knee	37	45	33	28	41	42	38	6	—
Popliteus	Knee	25	—	—	19	36	20	25	8	—
Gastrocnemius lateralis	Knee	30	42	30	20	23	34	30	8	—
Gastrocnemius medialis	Knee	20	16	29	22	20	32	23	6	—
Vastus medialis	Knee	36	51	48	50	30	44	43	9	—
Vastus intermedius	Knee	44	41	41	56	29	47	43	9	—
Vastus lateralis	Knee	24	45	42	56	36	50	42	11	—
Rectus femoris	Hip/Knee	63/47	47/43	33/54	22/56	62/31	76/43	50/46	20/9	49/—
Sartorius	Hip/Knee	61/22	68/14	58/23	46/29	65/8	79/27	63/21	11/8	55/—
Gracilis	Hip/Knee	41/16	74/16	79/26	71/31	80/22	86/27	72/23	16/6	72/—
Semitendinosus	Hip/Knee	51/7	44/14	57/31	64/21	18/16	31/19	44/18	17/8	57/—
Semimembranosus	Hip/Knee	38/26	29/24	65/19	44/36	21/14	31/14	38/22	15/8	46/—
Biceps femoris, caput longus	Hip/Knee	45/38	37/47	66/37	57/28	22/44	29/43	43/40	17/7	58/—
Tensor fasciae latae	Hip/Knee	27/22	42/36	55/36	41/32	57/37	47/31	45/32	11/6	65/—

from three-dimensional attachment areas rather than estimated in planar models.

The absolute volumes of formalin-fixed muscles reported in Table 2 may not be representative of those of living subjects, but they give an idea of the relative variability of the sample. Muscle volumes determined by Jensen and Davy (1975) for the m. gluteus medius (259–435 ml), m. rectus femoris (141–201 ml) and m. sartorius (108–173 ml) are comparable to those reported here in Table 2. The variation in the total muscle volumes in our study was considerable (2635–4155 ml), and may be due to differences in physical condition.

The wide anatomical variability observed similarly in this study and others suggest that a mean data set is artificial and may not be meaningful, biomechanically. Drawing conclusions based upon special features of these artificial anatomical configurations can be misleading.

To avoid drawing conclusions which are based on a special anatomical situation the sensitivity of any musculo-skeletal analysis (i.e. gait analysis, biomechanical testing involving muscles) to anatomical variability should be included in the analysis. The authors of this paper present a method to quantify the influence of anatomical variability on biomechanical modeling parameters using muscle moment arms as an example. In general, the moment arm magnitudes reported in this study (Table 4) are comparable to published data (i.e. Dostal *et al.*, 1986). However, the standard deviations of moment arms can be as large as $SD = 65\%$ in single muscles. Further, it is demonstrated in Table 4 that some muscle moment arms are very sensitive to the anatomical situation (i.e. m. gluteus maximus, m. rectus femoris) whereas others are not (i.e. m. gluteus medius, m. gastrocnemius).

This illustrates that biomechanical models which are used to investigate musculo-skeletal relationships should respect attachment location variability. Furthermore, models dealing with loads transferred from soft tissue structures to bones should consider the inter-individual variability in size and shape of attachment areas.

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